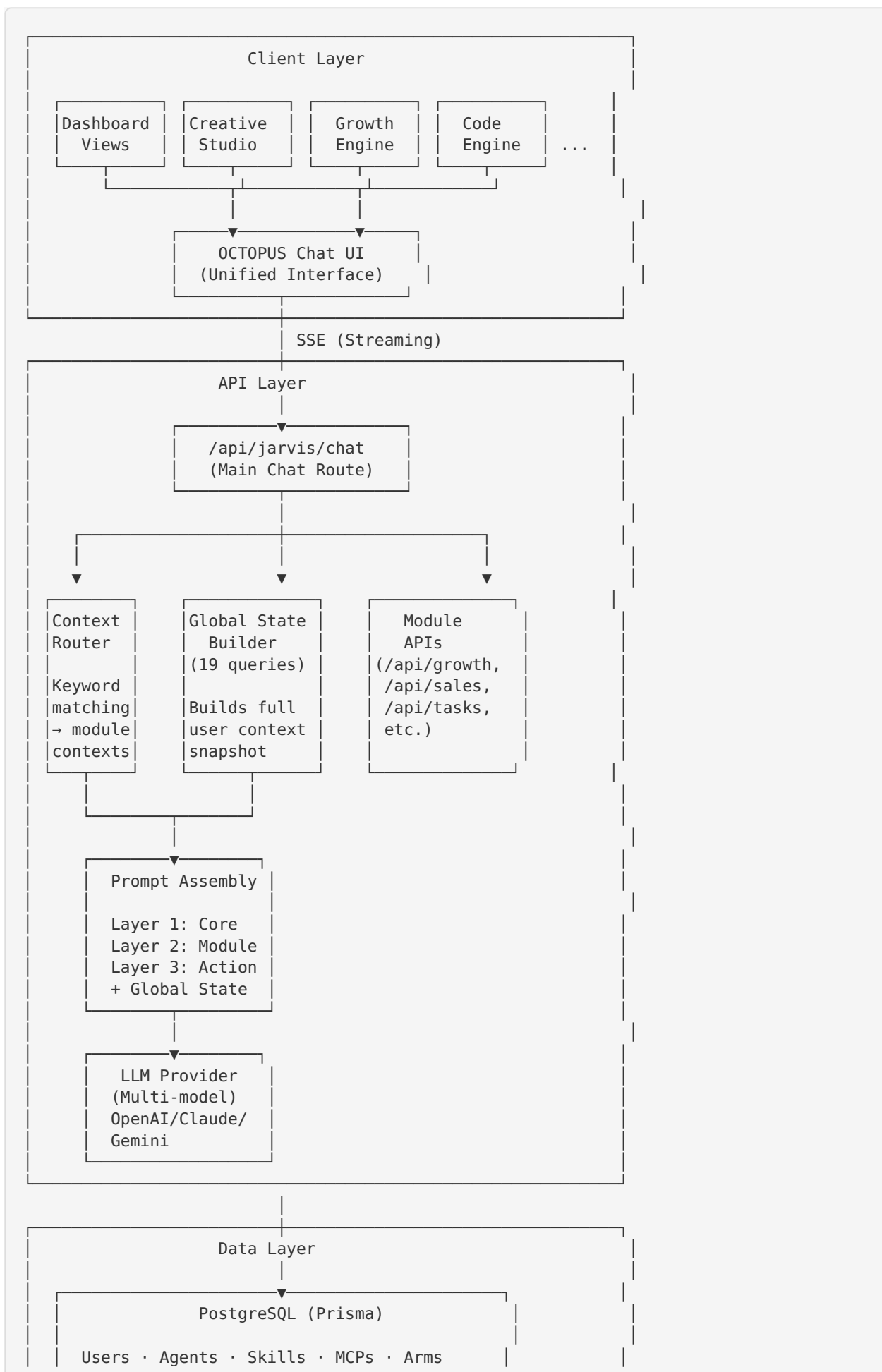
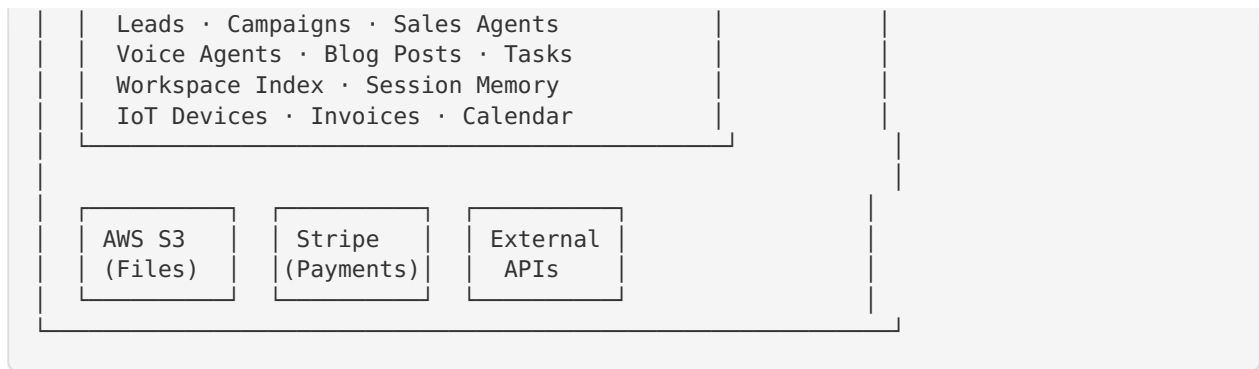


Octopus AI — Architecture Document

System Architecture

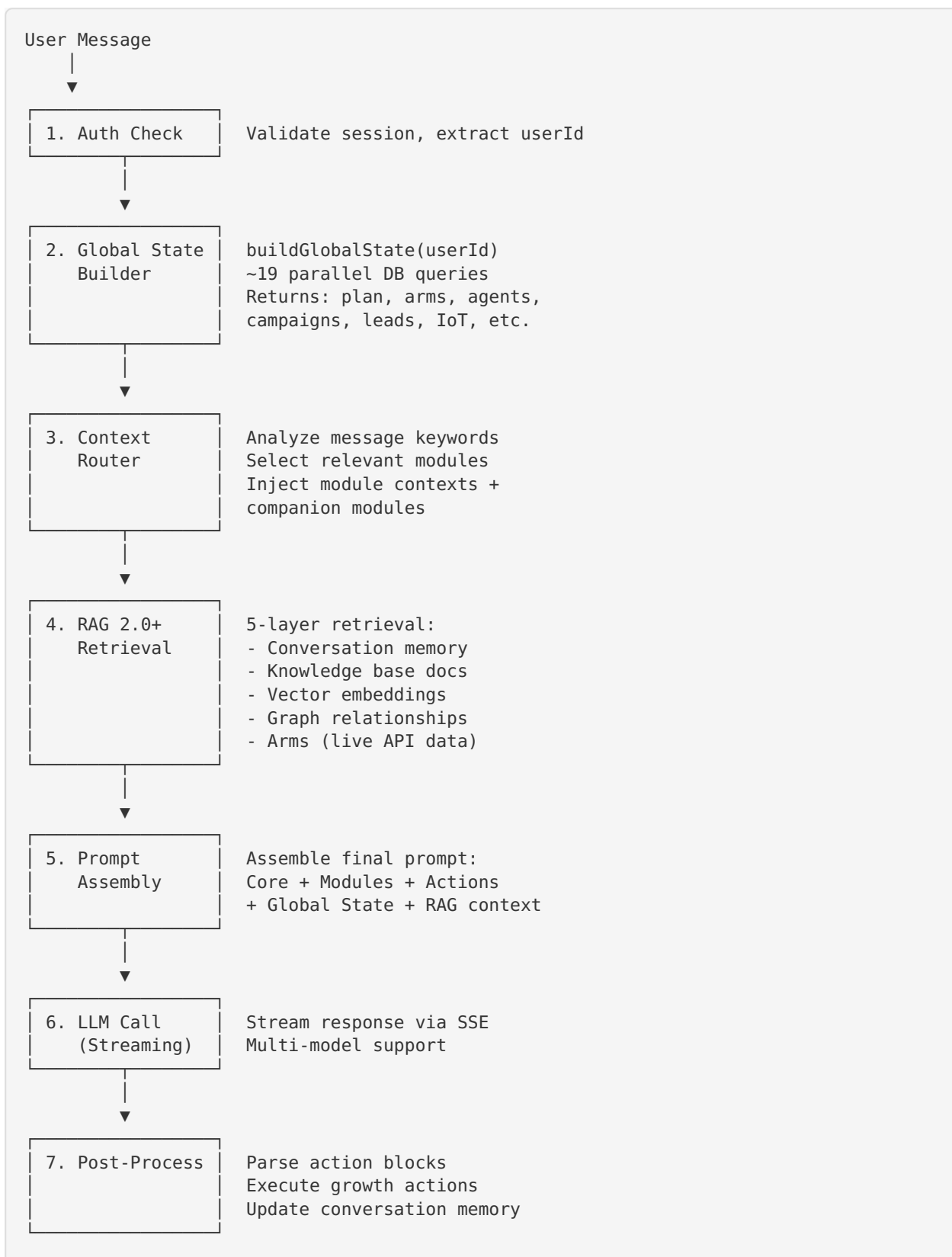
Octopus AI follows a **modular monolith** architecture where all modules share a common runtime but maintain clear boundaries through dedicated API routes, prompt contexts, and data models.





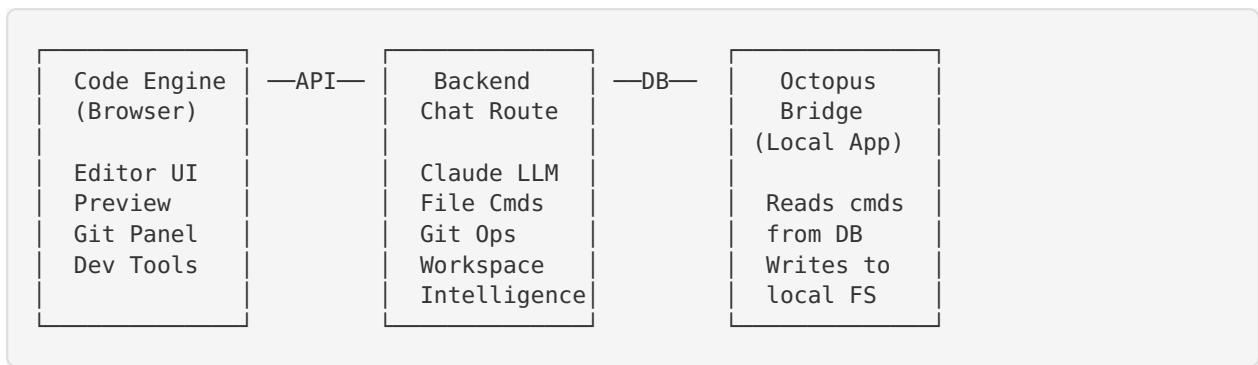
OCTOPUS Agent Flow

When a user sends a message, the following pipeline executes:



Code Engine Architecture

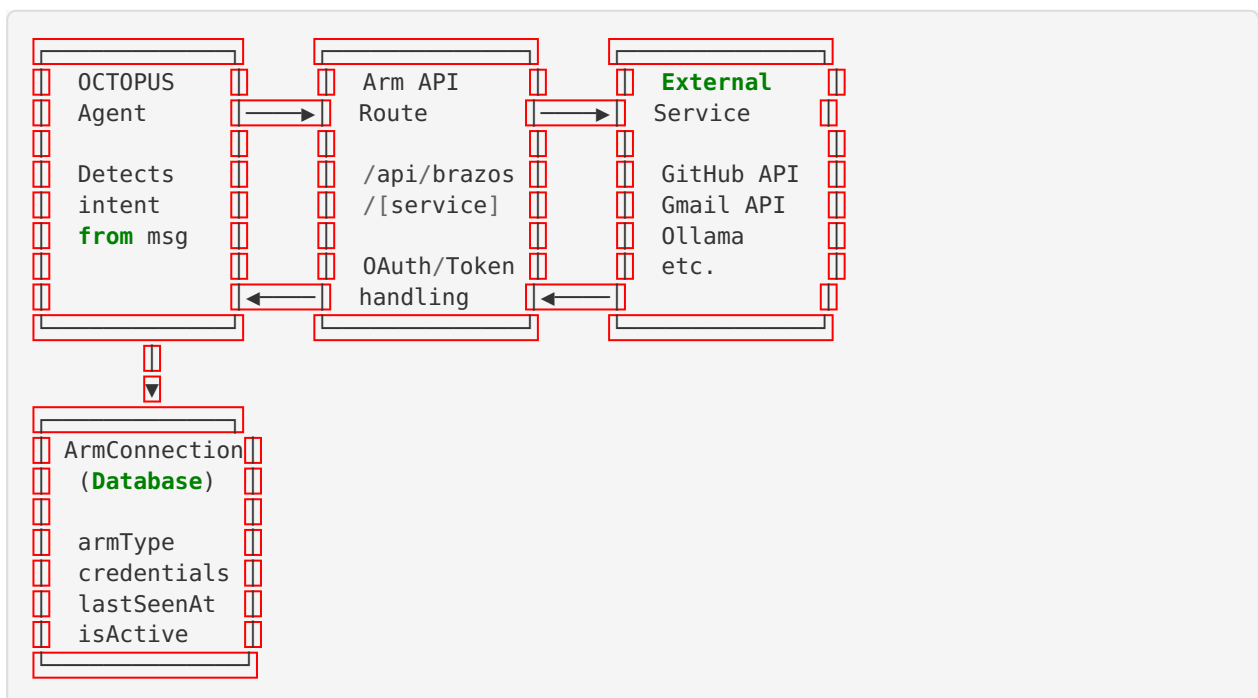
The Code Engine operates with a unique **Bridge** pattern:



- **Session-based**: Each coding session has a unique ID tracking files and changes
- **Bridge Protocol**: Commands (`write_file`, `execute_cmd`) stored in DB, polled by Bridge app
- **Workspace Intelligence**: `WorkspaceIndex` model tracks all files, `FileChangeEvent` tracks changes
- **Template System**: 6 starter templates with category filtering and tech stack badges

Integration Architecture (Brazos)

External integrations follow a consistent pattern:



Each Brazo (Arm) is registered as an `ArmConnection` in the database with:

- **armType**: Service identifier (github, gmail, telegram, ollama, etc.)
- **Credentials**: Encrypted OAuth tokens or API keys
- **Status Tracking**: `lastSeenAt` for connection health monitoring

Security Model

- **Authentication**: NextAuth.js with credential-based + OAuth providers
- **Authorization**: Session-based, all API routes validate `getServerSession(authOptions)`
- **Data Isolation**: All queries filtered by `userId` — users can only access their own data
- **Encryption**: AES-256 for sensitive data at rest, HTTPS for transit

- **API Keys:** Stored in environment variables, never exposed client-side
- **Rate Limiting:** Per-user limits based on subscription plan

Blog Architecture

Two blog systems run within the platform:

1. **Octopus Blog** (octopuskills.com/blog)
 - Direct DB writes to `OctoBlogPost` table
 - Managed by MAGUI agent via `octopus_seo_publish` skill
 - Auto-publish cron at scheduled intervals
 - SEO-optimized with structured data, sitemap integration
2. **Wildverse Blog** (external)
 - Published via API to external endpoint
 - Managed by VERA agent via `wildverse_seo_publish` skill
 - Separate deployment with its own database

Deployment

- **Production:** octopuskills.com (custom domain)
- **Build:** Standalone serverless output
- **Database:** Shared PostgreSQL instance (dev + prod)
- **CDN:** Static assets served via S3 + CloudFront